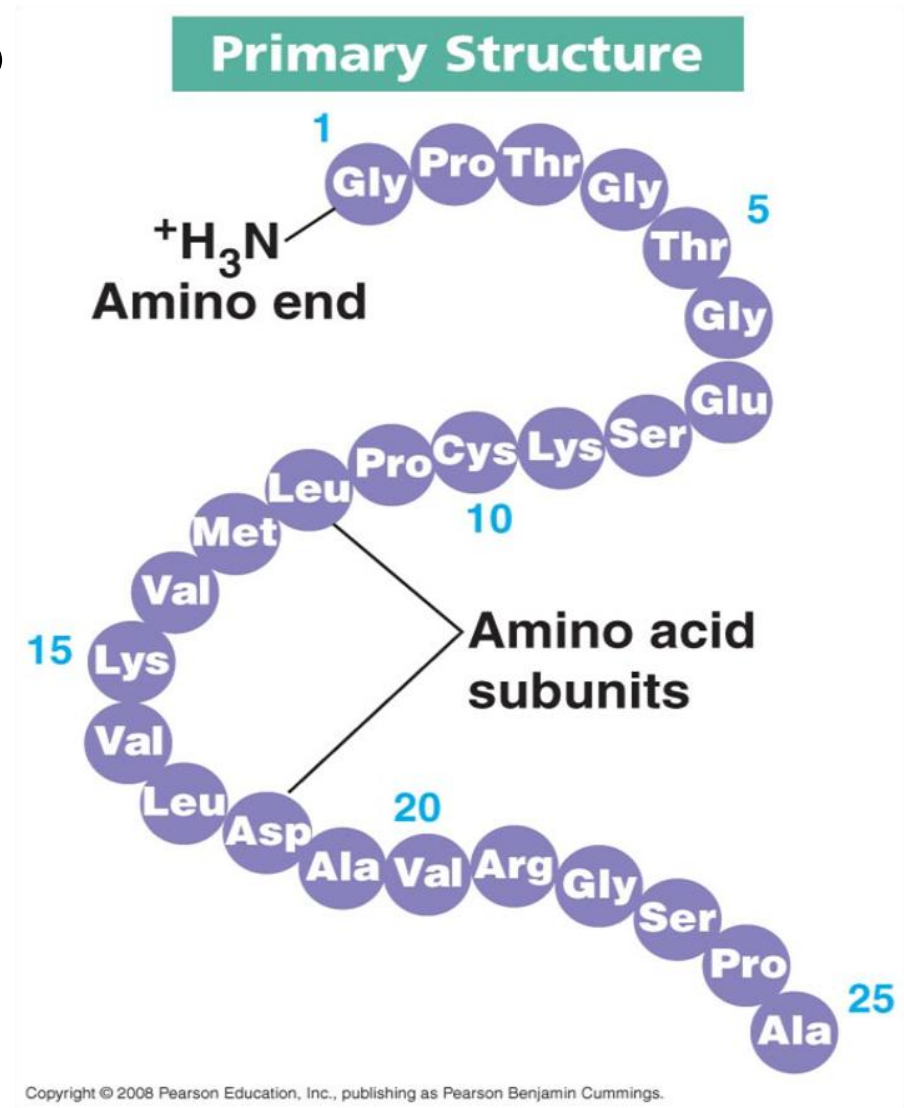


PROTEIN STRUCTURES

There are **3** main protein structures i.e. ***primary structure***, **secondary structure** and **tertiary structure**.

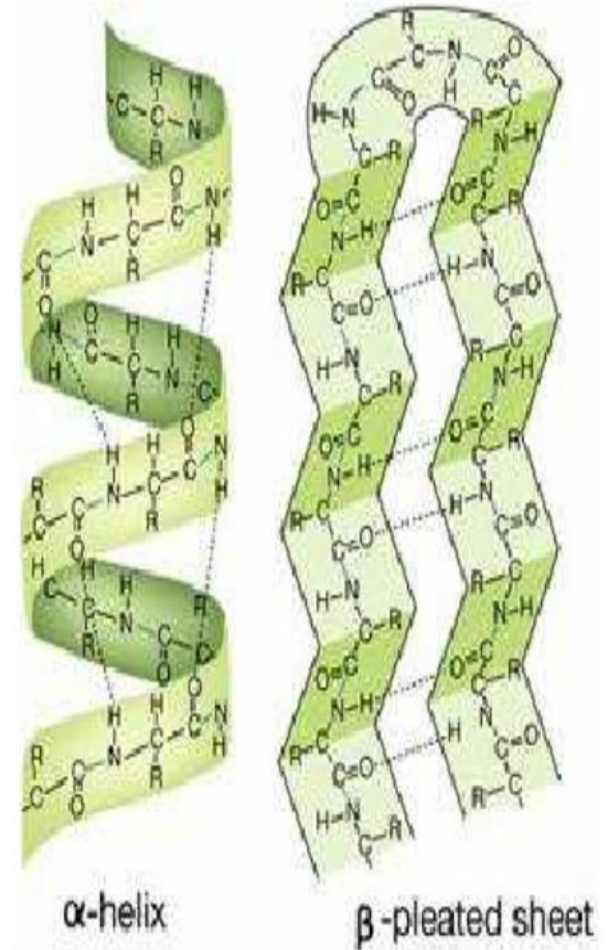
a) Primary structure

- ❑ It is a sequence of amino acids in a polypeptide chain.
- ❑ It is made up of 2 polypeptide chains held together by di-sulphide bridges.
- ❑ The sequences of amino acids of a protein dictate its biological functions.
- ❑ Examples of primary structures are ***insulin*** and ***lysosomes***.



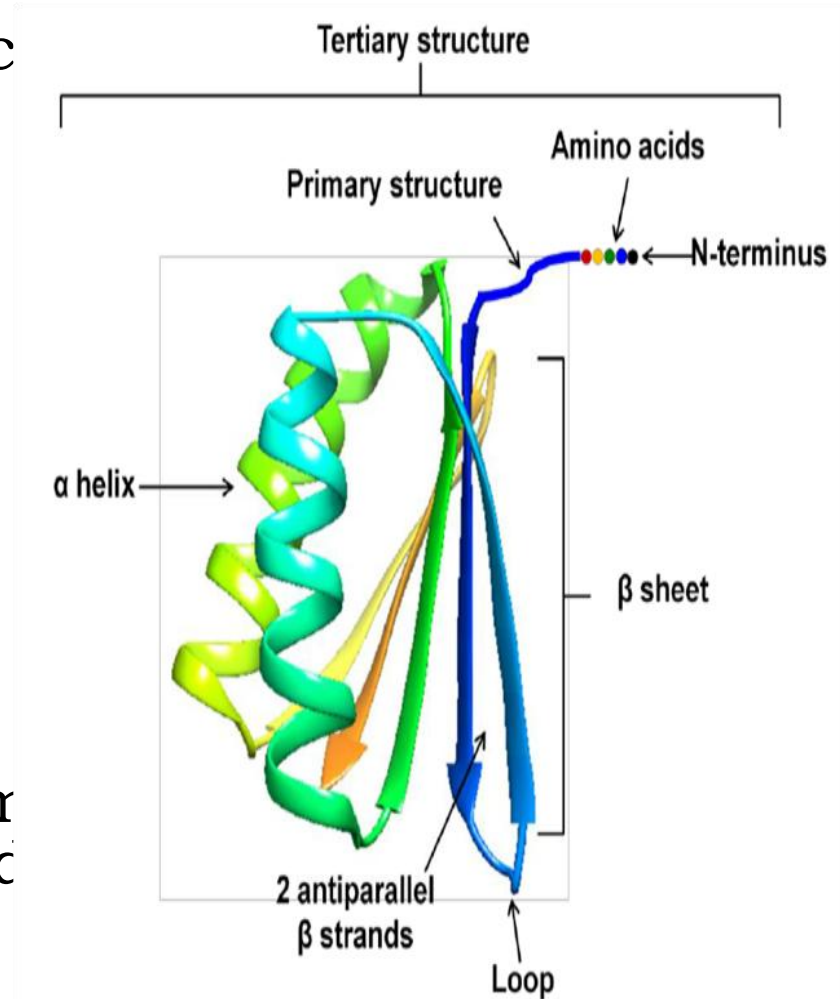
b) Secondary structure

- It is maintained by many ionic bonds which are formed between neighbouring **COO⁻** and **NH₃⁺** groups.
- The **hydrogen atoms** of the NH₃⁺ group of one amino acid is bonded to the oxygen atom of the COO⁻ group of another amino acid. Such structures are found in vertebrate skins, horns, nails, claws, beaks and feathers.
- Its hardness and stretch ability vary with a degree of cross linkage between di sulphide bridges and neighbouring chains. They are joined together by hydrogen bonds between adjacent chains.



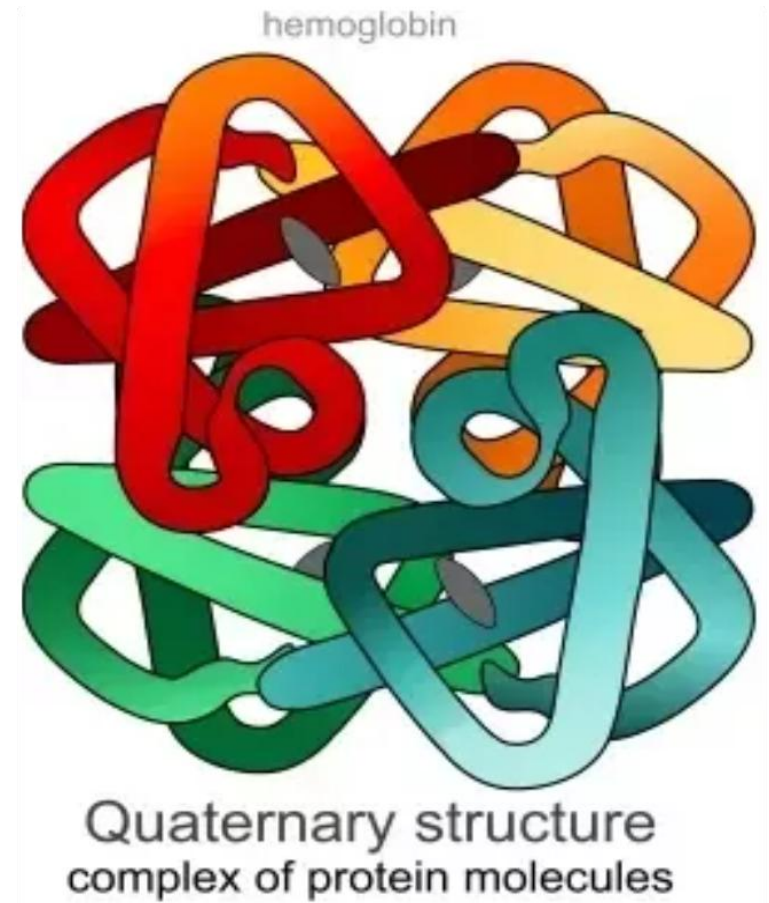
c) Tertiary structures

- ❖ The polypeptide chain coils extensively forming a compact **globular shape**. This structure is maintained by interaction of the four types of bonds i.e. ionic, hydrogen, disulphide bonds and hydrophobic interactions.
- ❖ The **hydrophobic interactions** are quantitatively the most important and occur when a protein folds to shield the hydrophobic side groups from the aqueous surrounding and at the same time exposing hydrophobic side chains.



d) Quaternary structure

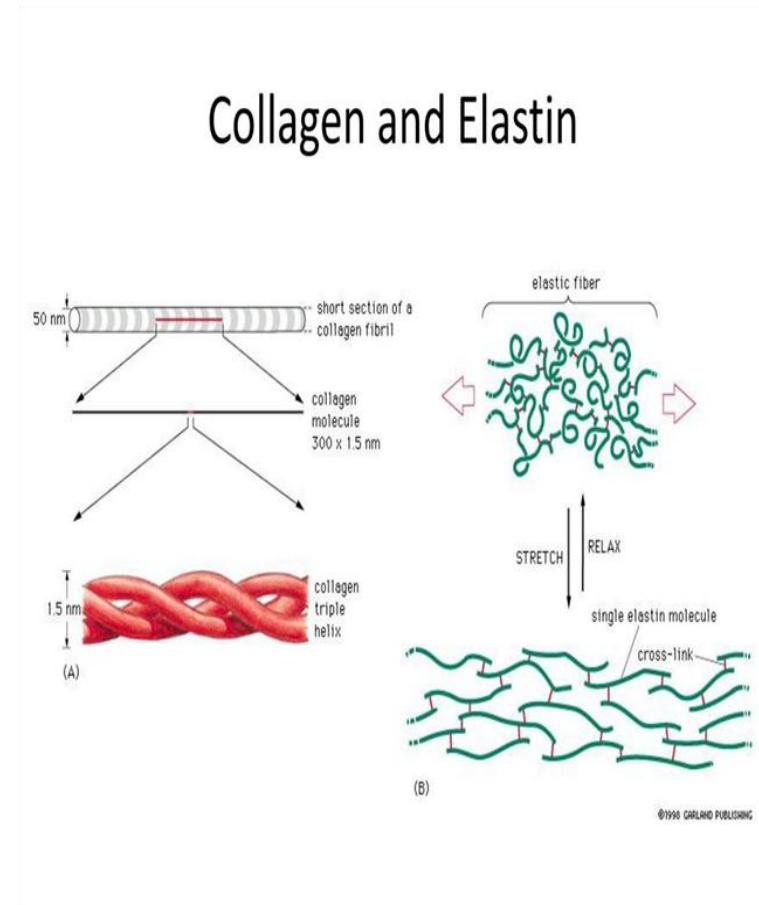
It is a combination of ***several polypeptide chains*** clumped together and associate with non-protein parts to form complex proteins e.g. in **haemoglobin**.



TYPES OF PROTEINS

1. Fibrous proteins (plays structural roles)

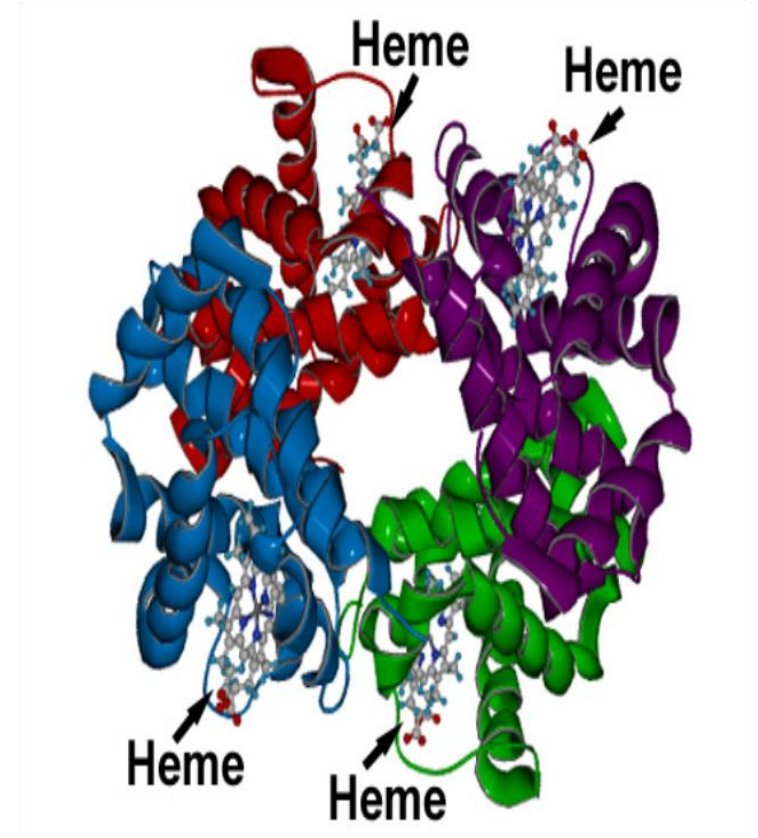
- ❖ These form **long chains** which may run parallel to one another being linked by cross bridges. They are very stable molecules and have structural roles within the organism e.g. collagen is made of such proteins.
- ❖ It has a primary structure which is a repeat of tripeptide sequence (glycine, proline and alanine) and forms a long unbranched chain.



TYPES OF PROTEINS

2. Globular proteins (plays metabolic roles)

- ❖ They have a highly irregular sequence of amino acids in their polypeptide chains. Their shape is compact and globular. All enzymes are globular proteins.
- ❖ Others include hormones and haemoglobin.

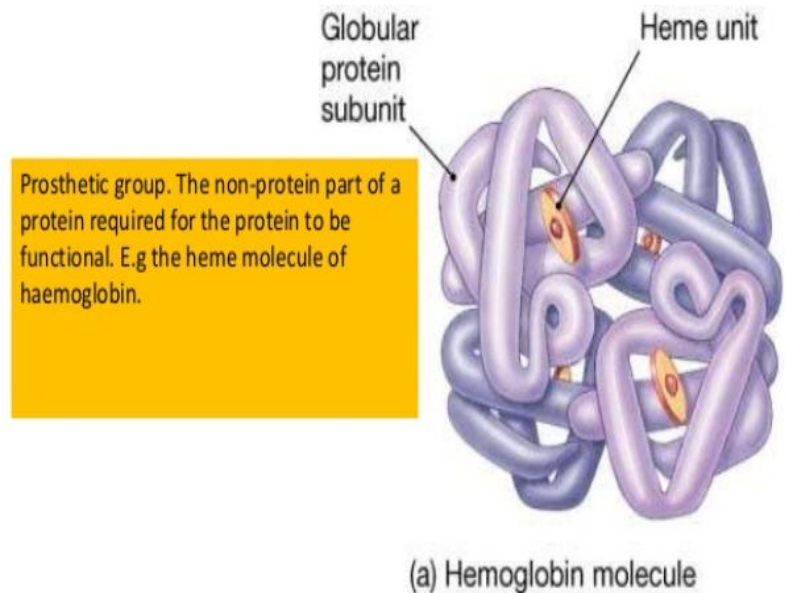


TYPES OF PROTEINS

3. Conjugated proteins

- ❖ These are proteins which incorporate other chemicals within their structure.
- ❖ The ***non-protein*** part is the ***prosthetic group*** and plays a virtual role in the functioning of the proteins

7.5.1: Conjugated Protein



Conjugated proteins

Name of protein	Where it is found	Prosthetic group
Haemoglobin	Blood	Haem (iron)
Mucin	Saliva	Carbohydrate
Casein	Milk	Phosphoric acid
Cytochrome oxidase	Electron carrier path way	Copper
Nucleoplasm	Ribosomes	Nucleic acid

QUESTION:

***HOW DOES THE MOLECULAR
STRUCTURE OF PROTEINS RELATE
TO THEIR ROLES?***

HOW DOES THE MOLECULAR STRUCTURE OF PROTEINS

RELATE TO THEIR ROLES?

- ❖ Some proteins have a **structural function**, these are fibrous proteins with a secondary structure insoluble in water and physically tough e.g. **collagen** in connective tissues, bone, tendons and cartilage. Other structural proteins include **keratin** in feathers, nails, hair, horns, beaks and skin.
- ❖ Some proteins function as **enzymes**. These have a globular structure and are soluble in water e.g. **digestive enzymes** like pepsin, respiratory and photosynthetic enzymes.
- ❖ Some proteins function as **hormones** regulating metabolic **processes**. These are globular and soluble in water e.g. insulin which regulates metabolic activity.
- ❖ Some proteins function as **respiratory pigment**. These are **globular proteins** with a quaternary structure that increases their surface area for **transport** or **storage** of respiratory gases e.g. haemoglobin which transports oxygen in blood and **myoglobin** that stores oxygen in muscles.

HOW DOES THE MOLECULAR STRUCTURE OF PROTEINS RELATE TO THEIR ROLES?

- ❖ Some proteins are involved in **transport** and are globular with primary or tertiary structures e.g. **serum albumen** that transports fatty acids and lipids in blood.
- ❖ Some proteins are involved in **immunological responses** hence protecting the body. These are globular e.g. **antibodies, fibrinogen** and thrombin.
- ❖ Some proteins are **contractile** e.g. they are fibrous with a secondary structure e.g. myosin and actin **filaments in muscles**.
- ❖ **Storage proteins** are toxins and soluble and water with a globular structure e.g. snake venom, bacteria toxins, etc

HOW DOES THE MOLECULAR STRUCTURE OF PROTEINS RELATE TO THEIR ROLES?

- ❖ Some proteins are ***insoluble in water*** e.g. **ovalbumin** that occurs in **egg white**, **casein** in milk, etc.
- ❖ Globular proteins form ***colloidal suspensions*** that hold molecules in position within cells e.g. proteins in the cytoplasm of most cells where they are soluble in water and have a large surface area.
- ❖ ***Globular proteins*** in blood are **buffers** since they are soluble in water.

DENATURATION OF PROTEINS

- The **three dimensional structure** of the protein is due to weak ionic and hydrogen bonds.
- Any agent which **breaks** these **bonds** causes the three dimensional shape to be changed to a more fibrous form. This process is known as **denaturation**.
- In case the actual sequence of the amino acid is not altered but only the **overall shape** of the molecule is **changed**.

Factors causing protein denaturation

Factor	Explanation	Example
1. Heat	Causes the atoms of the protein to vibrate more thus breaking the hydrogen and ionic bond.	Coagulation of albumen (egg white becomes more fibrous).
2. Acids	Addition of hydrogen ions in acids combine with COO^- of amino acids and form COOH ionic bonds are hence broken.	Souring of milk by acid and lowering pH of casein making it insoluble.
3. Alkalis	Reduced number of H^+ cause NH^+ group to loose H^+ to form NH_2 therefore ionic bonds broken.	Souring of milk by alkalis.

Factors causing protein denaturation

Factor	Explanation	Example
4. Inorganic chemicals	Ions of heavy metals e.g. mercury and silver combine with COO^- groups destructing the ionic bonds.	<i>Enzymes are inhibited by being destructed in presence of ions e.g. cytochrome oxidase.</i>
5. Organic chemicals	Organic solvents alter hydrogen bonding with a protein.	<i>Alcohol denatures certain bacterial proteins. This is what makes it useful for sterilization.</i>
6. Mechanical force	Physical movement can break hydrogen bonds.	<i>On stretching a hair, the hydrogen bonds in the keratin helix is extended and hair stretches.</i>

Functions of proteins

VITAL ACTIVITY	PROTEIN EXAMPLE	FUNCTION
Nutrition	Digestive enzymes e.g. trypsin, amylase, etc. Fibrous proteins in grana lamellae casein	<i>catalyses, hydrolysis of proteins to peptides.</i> <i>Helps to arrange chlorophyll molecules to receive unlimited light.</i> <i>Assists in transporting of food in filter feeder.</i> <i>Storage of proteins in milk.</i>
Respiration and transport.	Haemoglobin. Myoglobin Prothrombin/fibrinogen Antibodies.	<i>Transport of oxygen.</i> <i>Stores oxygen in muscles.</i> <i>Required for blood clotting.</i> <i>Essential for defense.</i>

Functions of proteins

VITAL ACTIVITY	PROTEIN EXAMPLE	FUNCTION
• Excretion	Enzymes e.g. urease	<i>Catalyzes reaction in ornithine cycle and helps in protein break down and urea formation</i>
• Support and movement	Actin/myosin	<i>Makes it easy for muscle contraction.</i>
	Collagen	<i>Gives strength with flexibility in tendons and cartilage.</i>
	Keratin	<i>Tough for protection e.g. in scales, claws, nails, hooves, etc.</i>
	Sceleratin	<i>Provide strength in insect exo-skeleton</i>

Functions of proteins

VITAL ACTIVITY	PROTEIN EXAMPLE	FUNCTION
• Growth	Hormones e.g. thyroxine	<i>Controls growth and metabolism.</i>
• Sensitivity and co-ordination.	Hormones e.g. insulin	<i>Control of blood sugars</i>
	Vasopressin	<i>Control of blood pressure</i>
	Rhodopsin	<i>Visual pigments in retina.</i>
• Reproduction	Hormones e.g. prolactin	<i>Induces milk production in mammals.</i>
	Chromatin	<i>Gives structural support to chromosomes.</i>
	Chitin	<i>Storage of proteins in seeds which nourishes the embryo.</i>
	Keratin	<i>Forms horns and antlers which are used for sexual display.</i>